

Technical Document #36: Data Engineering - Comprehensive Overview

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Schema evolution handles changes in data structure over time. Schema registries manage schema versions and ensure backward compatibility.

Data lineage tracks the origin and movement of data through systems. It is essential for debugging, compliance, and impact analysis.

ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loads it into a target data store.

Data pipelines automate the movement and transformation of data between systems. Apache Airflow and Prefect are popular workflow orchestration tools.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Key Takeaways:

- Data lakes store raw data in its native format.
- Partitioning divides data into smaller, more manageable pieces.
- Data pipelines automate the movement and transformation of data between systems.